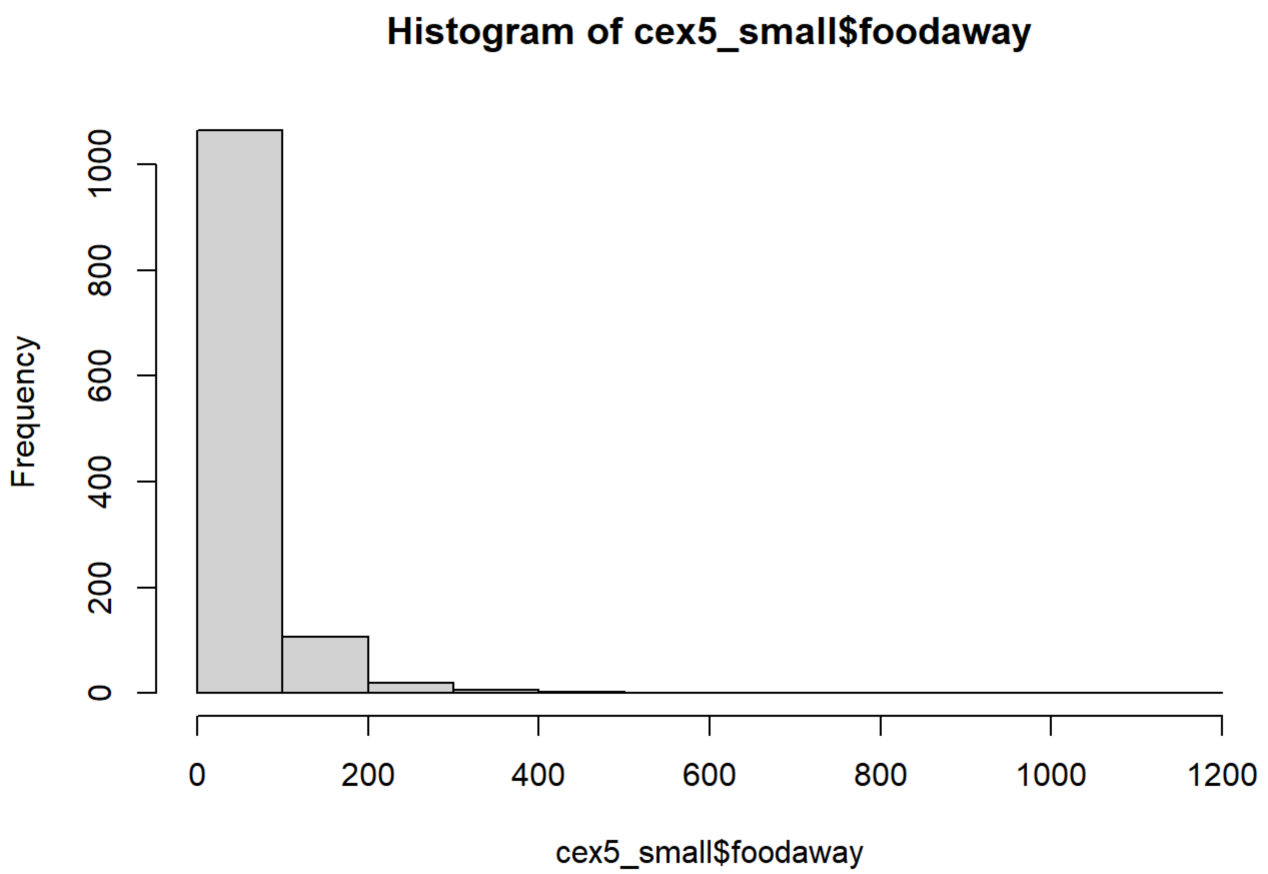
HW2.25

(a)

```
cex5_small <- read.csv("cex5_small.csv")
hist(cex5_small$foodaway)
```




```
summary(cex5_small$foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   12.04   32.55   49.27   67.50 1179.00
```

- Mean = 49.27
- Median = 32.55
- 25th percentile = 12.04
- 75th percentile = 67.50

(b)

```
# Advanced degree households
mean(cex5_small$foodaway[cex5_small$advanced == 1])
```

```
## [1] 73.15494
```

```
median(cex5_small$foodaway[cex5_small$advanced == 1])
```

```
## [1] 48.15
```

```
# College degree households
mean(cex5_small$foodaway[cex5_small$college == 1])
```

```
## [1] 48.59718
```

```
median(cex5_small$foodaway[cex5_small$college == 1])
```

```
## [1] 36.11
```

```
# No college or advanced degree
mean(cex5_small$foodaway[cex5_small$college == 0 & cex5_small$advanced == 0])
```

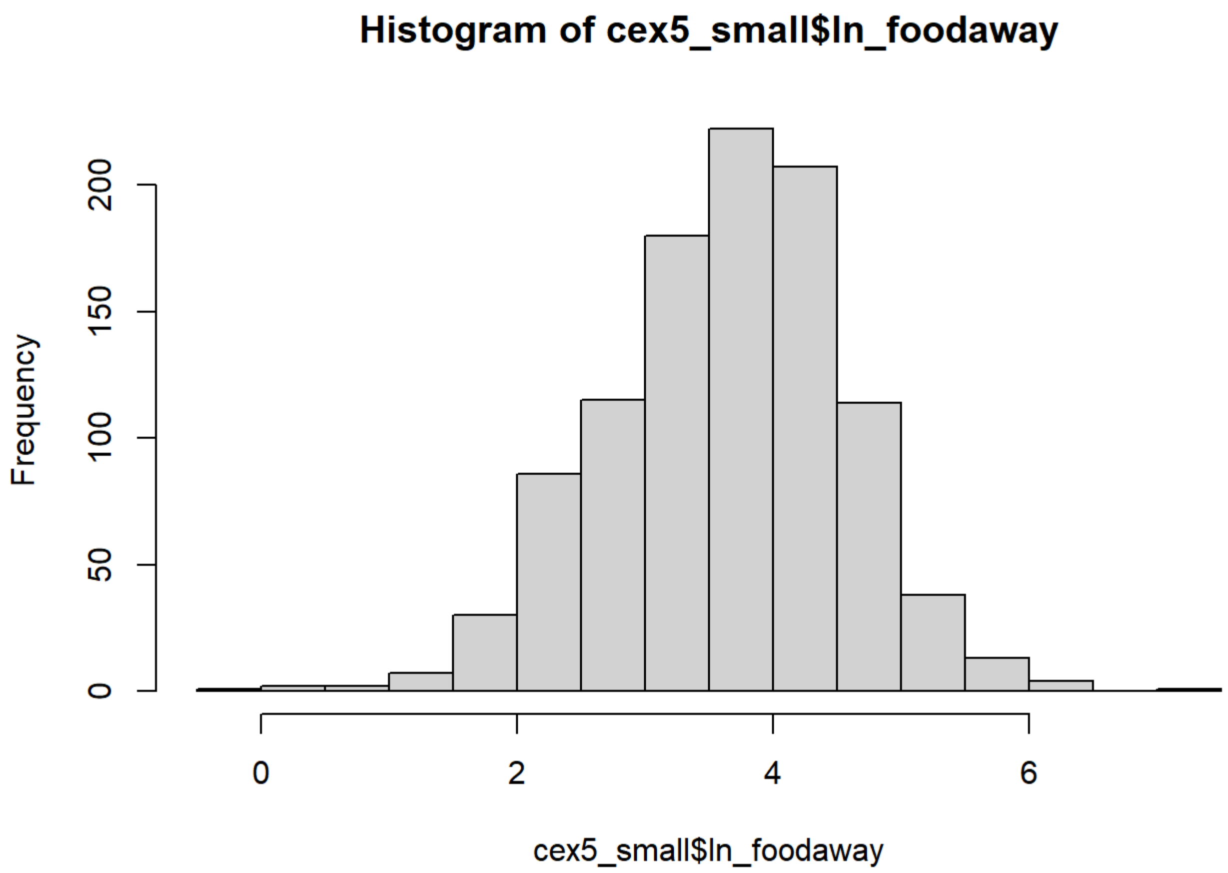
```
## [1] 39.01017
```

```
median(cex5_small$foodaway[cex5_small$college == 0 & cex5_small$advanced == 0])
```

```
## [1] 26.02
```

(c)

```
cex5_small$ln_foodaway <- log(cex5_small$foodaway)
hist(cex5_small$ln_foodaway)
```




```
cex5_small$ln_foodaway[cex5_small$ln_foodaway == -Inf] <- NA
summary(cex5_small$ln_foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
## -0.3011  3.0759  3.6865  3.6508  4.2797  7.0724     178
```

FOODAWAY 和 ln(FOODAWAY) 的觀測值數量不同，是因為有些 FOODAWAY 的數值等於 0，而 0 的自然對數是不存在的。因此在計算 ln(FOODAWAY) 時，這些觀測值無法取對數，會被視為缺失值（NA=178）而被排除，所以 ln(FOODAWAY) 的有效觀測數會比 FOODAWAY 少。

(d)

```
model <- lm(ln_foodaway ~ income, data = cex5_small)
summary(model)
```

```
##
## Call:
## lm(formula = ln_foodaway ~ income, data = cex5_small)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
## income      0.0069017  0.0006546   10.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
##   (因為不存在，178 個觀察量被刪除了)
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

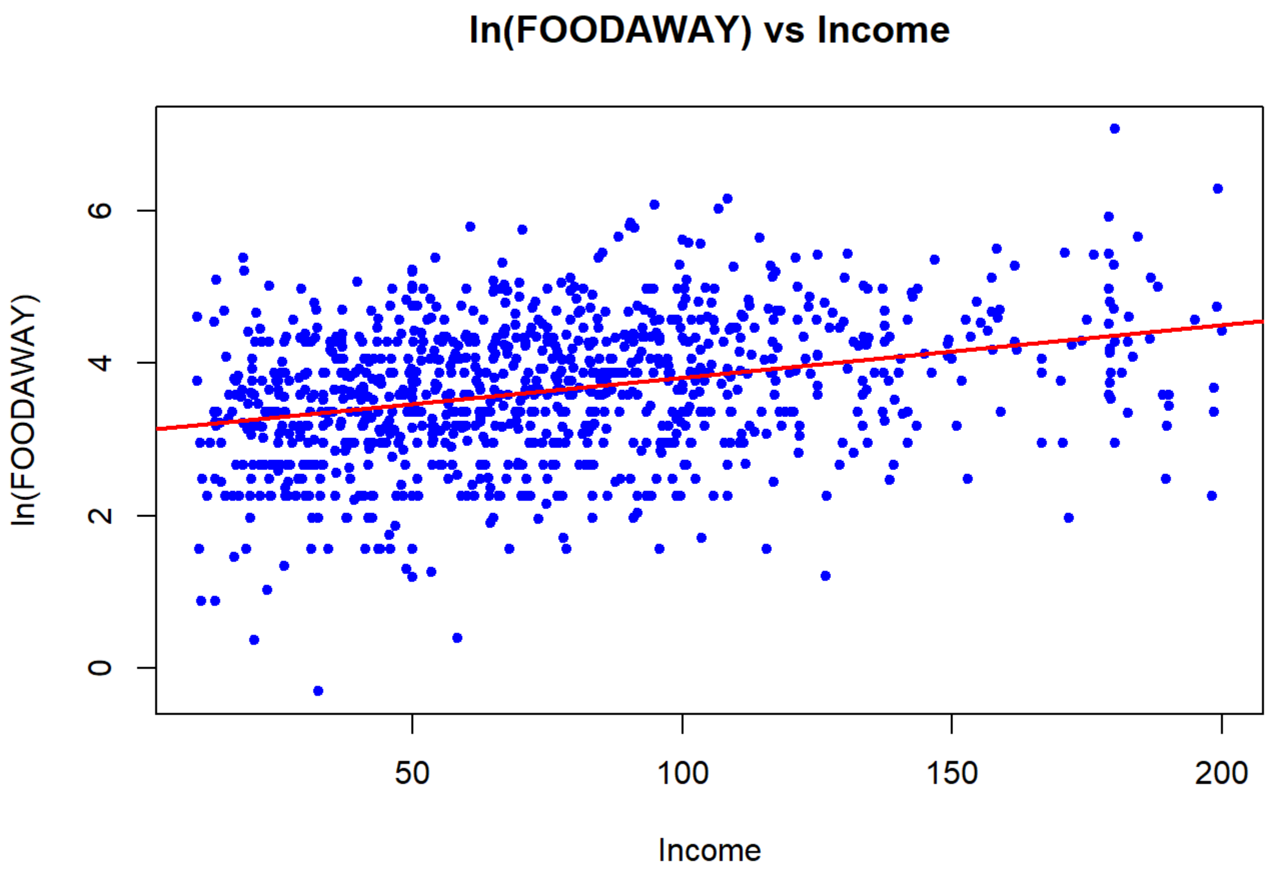
其餘條件不變，當家庭月收入增加 $100 時，每人外食支出平均約增加 **0.69%**

此外，該係數的 p-value 小於 0.001，表示此關係在統計上高度顯著。這說明收入越高的家庭，通常會有較高的外食支出。

(e)

```
plot(cex5_small$income, cex5_small$ln_foodaway,
     xlab = "Income",
     ylab = "ln(FOODAWAY)",
     main = "ln(FOODAWAY) vs Income",
     pch = 20,
     col = "blue")

abline(model, col = "red", lwd = 2)
```

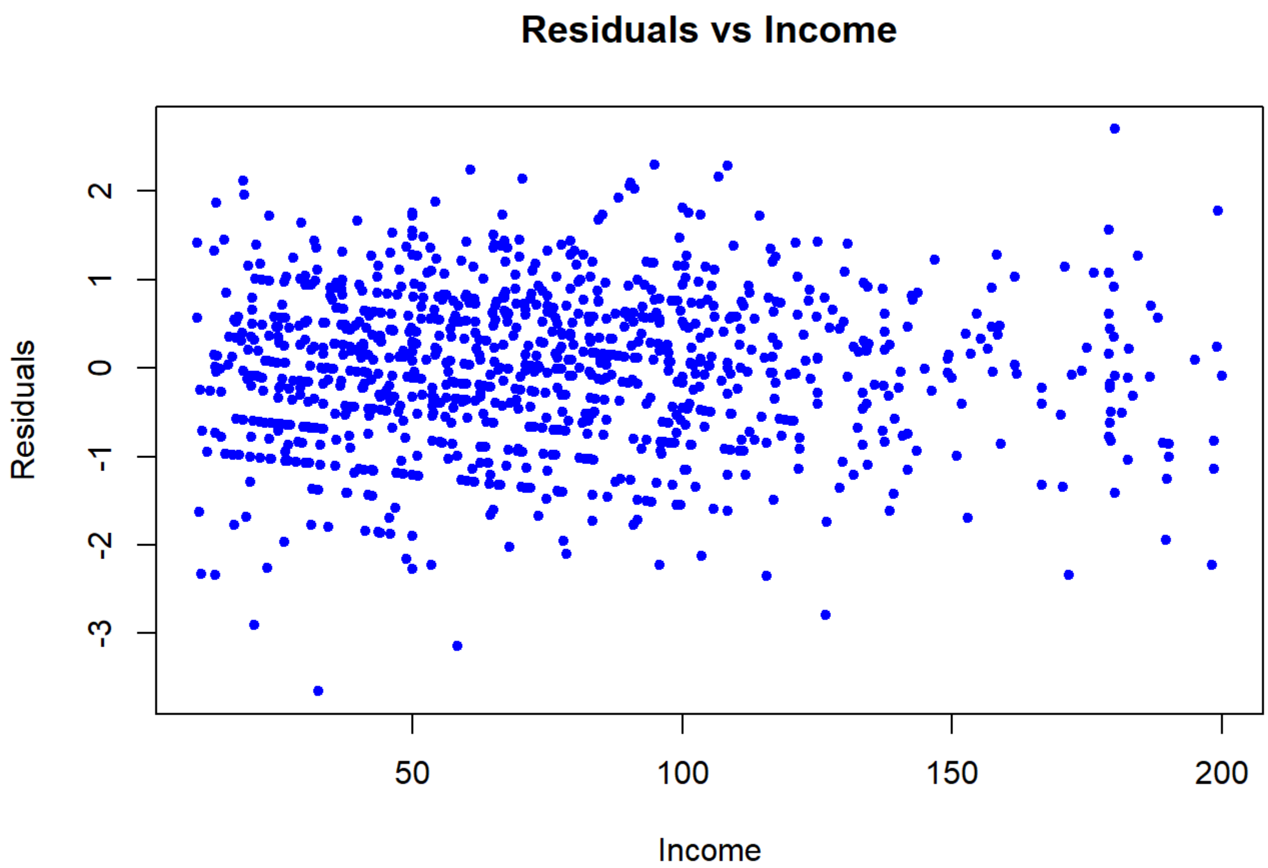


散佈圖顯示 ln(FOODAWAY) 與 INCOME 之間存在正向關係，回歸線向上傾斜，表示收入較高的家庭通常會有較高的外食支出。

(f)

```
residuals_d <- residuals(model)

plot(model$model$income, residuals(model),
     xlab = "Income",
     ylab = "Residuals",
     main = "Residuals vs Income",
     pch = 20,
     col = "blue")
```



殘差圖顯示殘差大致隨機分布在 0 附近，沒有出現明顯的系統性模式，例如曲線形狀或明顯的趨勢。這表示模型的線性形式是合理的，收入與 ln(FOODAWAY) 之間的線性關係基本上可以描述資料